

MM-ARC: Multimodal Adaptive Routing of Capital with Robustness-Audited Strategy Pools

Yang Chen
Zhejiang University
Hangzhou, China
ychen2014@zju.edu.cn

Yuchen Cao
City University of Hong Kong
Hong Kong, China
cynthcao2-c@my.cityu.edu.hk

Jacky Keung
City University of Hong Kong
Hong Kong, China
jacky.keung@cityu.edu.hk

Leilei Gan
Zhejiang University
Hangzhou, China
leileigan@zju.edu.cn

Kun Kuang
Zhejiang University
Hangzhou, China
kunkuang@zju.edu.cn

Yueheng Jiang
Zhejiang University
Hangzhou, China
yuehengj@zju.edu.cn

Zhaozhao Ma
Zhejiang University
Hangzhou, China
zhaozhaoma@zju.edu.cn

Jianping Zhu
Dalian University of
Technology
Dalian, China
zhujp@mail.dlut.edu.cn

Fei Wu
Zhejiang University
Hangzhou, China
wufei@zju.edu.cn

Jinpeng Li*
Zhejiang University
Hangzhou, China
lijinpeng@zju.edu.cn

Abstract

Financial trading systems must convert multimodal market history into executable positions while limiting overfitting from repeated strategy search. We introduce MM-ARC (MultiModal Adaptive Routing of Capital), which routes capital across trend, reversal, breakout, and exposure-control experts using aligned chart, numerical, and technical-text views. Within each market, regime-conditioned strategy pools are shared with bounded asset-specific adjustments. Robustness-Audited Bayesian Optimization (RABO) filters candidates proposed by Bayesian optimization on purged validation blocks using after-cost benchmark exceedance, lower-tail performance, stability, and turnover; a common portfolio layer then produces market-feasible orders. We evaluate 62 instruments across five asset classes using five training seeds and a frozen July 2025–June 2026 trading holdout. Under an all-in one-way cost of 10 basis points per unit of executed turnover, MM-ARC attains an equal-market Sharpe ratio of 1.33 and maximum drawdown of -13.7% , versus 0.53 and -18.3% for the LLMoE-style routing baseline. The global learned-static control reaches 1.12 and -15.3% , respectively. Paired block-bootstrap intervals favor the prespecified contrasts, while ablation point estimates are consistent with contributions from visual inputs, adaptive routing, exposure control, and robustness-audited admission. Family-level data-snooping tests also reject their prespecified nulls (SPA $p = .039$; Reality Check $p = .021$); we therefore interpret the evidence as benchmark-relative support within the evaluated candidate family and holdout, not as universal or future-regime superiority.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Economics*.

Keywords

multimodal learning, capital routing, robust strategy selection, financial trading, backtest overfitting

1 Introduction

Financial markets are non-stationary decision environments in which useful evidence is distributed across price sequences, chart structure, and textual market summaries. A model that predicts direction accurately can nevertheless fail as a trading system when

its signals compete for capital, violate market constraints, or disappear after turnover and transaction costs. The central problem is therefore not perception alone, but the conversion of multimodal evidence into a synchronized, executable portfolio.

Financial language models and multimodal agents broaden the information interface of trading systems [21, 22, 24, 25]. Their forecasts and reasoning, however, neither allocate capital among incompatible trading behaviors nor reconcile their target exposures under a common portfolio budget. Ordinary mixture-of-experts routing selects representations or computation paths; a trading router must instead allocate capital, with direct consequences for exposure, turnover, and drawdown [6, 13].

Adaptation creates an opposing risk. A fixed strategy pool is too rigid under regime shifts, but selecting the best Bayesian optimization (BO) backtest or accepting unconstrained LLM-generated strategies enlarges the effective hypothesis space and amplifies data snooping [2, 3, 8, 18]. A credible adaptive pool must therefore use only historically visible information and judge candidates by the distribution—including lower-tail and drawdown behavior—rather than the peak of their validation outcomes.

We propose MM-ARC, an execution-aware pipeline that connects multimodal perception, adaptive capital routing, interpretable strategy experts, robustness-audited search, and market-feasible portfolio construction. We evaluate the complete pipeline on point-in-time records for 62 instruments from five asset classes using a one-year frozen trading holdout and a common after-cost portfolio-construction layer. The paper makes three contributions:

- **Adaptive multimodal capital routing.** A shared encoder fuses aligned views, and a regime-conditioned router allocates capital over interpretable trend, reversal, breakout, and exposure-control experts before market-feasibility projection.
- **Market-shared expert strategy pools.** Instead of 744 asset-specific pools, MM-ARC uses 60 pools indexed by market, expert, and regime. Bounded, schema-valid asset-specific parameter adjustments (residuals) preserve strategy semantics and reduce repeated search.
- **Robustness-audited admission.** RABO combines benchmark exceedance, validation lower-tail performance, median behavior, local parameter stability, and turnover. It thereby trades validation peaks for stronger lower-tail and drawdown behavior.

*Corresponding author: Jinpeng Li (lijinpeng@zju.edu.cn).

Across five separately settled market portfolios under the normalized all-in 10-bps protocol, the arithmetic means of SR and MDD are 1.33 and -13.7% , versus .53 and -18.3% for the closest routing baseline. These are favorable holdout point differences of .80 in SR and 4.6 percentage points in MDD. SPA ($p = .039$) and Reality Check ($p = .021$) also reject their prespecified family-level nulls, providing benchmark-relative evidence after data-snooping correction on this holdout without implying dominance over every candidate or future regimes.

2 Related Work

Financial LLM agents. Financial LLMs and agents support text understanding, memory, forecasting, and multi-step trading workflows [19–24]. MM-ARC addresses a narrower portfolio problem: translating multimodal states into capital weights, target exposures, and feasible orders rather than producing open-ended analyses or executable code.

Multimodal forecasting. Financial models increasingly combine structured series, images, and language, while Time-LLM and Time-VLM-style methods adapt foundation models to temporal prediction [10, 12, 25]. Unlike systems that fuse independent news, filings, or macroeconomic text, MM-ARC uses visual and structured-text representations derived causally from the same observed OHLCV window and evaluates them by after-cost portfolio performance rather than forecasting accuracy alone.

Routing and expert trading. Modular trading systems and MoE models encourage specialization, but their routers commonly direct representations or select predictors [6, 13]. In MM-ARC, the router instead produces simplex-valued capital allocations over interpretable strategy experts and is evaluated through after-cost portfolio contributions, turnover, exposure, and regime-conditioned outcomes.

Backtest overfitting and robust search. Reality Check, Superior Predictive Ability (SPA), Probability of Backtest Overfitting (PBO), Deflated Sharpe Ratio, and System Parameter Permutation diagnose selection risk; Bayesian optimization and LABO guide search [2–4, 8, 14, 16–18]. RABO embeds past-only cross-block distribution audits in admission and capital weighting.

3 Problem Formulation

Let $m \in \mathcal{M}$ index a market and $u \in \mathcal{U}_m$ an asset. At decision time t , the model encodes only observations available by that cutoff:

$$o_{t,u} = (V_{t,u}, X_{t,u}, L_{t,u}), \quad z_{t,u} = \text{Enc}_\eta(o_{t,u}), \quad (1)$$

where the chart, temporal, and structured-text views share the same timestamp. This cutoff-aligned state prevents later records from entering the decision.

The capital router converts state into continuous expert weights. It combines $z_{t,u}$, market m , the supervised fine-tuning (SFT) prior $p_{t,u}$, and pool evidence $c_{m,\hat{r}_{t,u}}^{\text{rob}}$, where $\hat{r}_{t,u} = \arg \max_r p_{t,u,r}$ and $\eta_g \geq 0$ controls the evidence tilt:

$$w_{t,u} = \text{softmax}\left(f_\phi(z_{t,u}, m) + B_m p_{t,u} + \eta_g c_{m,\hat{r}_{t,u}}^{\text{rob}}\right) \in \Delta^K, \quad (2)$$

with $K = 4$. The simplex makes routing a capital reallocation among experts rather than a hard expert switch.

Every expert emits a target exposure $h_{t+1|t,u,k} \in [-1, 1]$ in the same unit: a fraction of market-portfolio NAV before routing. To make these targets tradable, their router-weighted signals are synchronized on the native market calendar and projected onto the feasible set C_m , which encodes position, leverage, short-sale, cash, and turnover limits:

$$v_{t+1|t,m,k} = \left(\frac{w_{t,u,k} h_{t+1|t,u,k}}{|\mathcal{U}_m|} \right)_{u \in \mathcal{U}_m},$$

$$h_{t+1|t,m}^{\text{raw}} = \sum_{k=1}^K v_{t+1|t,m,k}, \quad (3a)$$

$$h_{t+1|t,m}^* = \arg \min_{h \in C_m} \|h - h_{t+1|t,m}^{\text{raw}}\|_2^2 + \lambda_q \|h - h_{t,m}^*\|_1. \quad (3b)$$

Projection is performed separately for each market portfolio and reconciles both expert disagreement and cross-asset constraints. The order $q_{t+1,m} = h_{t+1|t,m}^* - h_{t,m}^*$ is submitted at the next tradable open, with cost and return booked to $t + 1$; no expert receives credit for exposure removed by projection.

Settlement deducts applicable commissions, slippage, taxes, roll costs, funding, and soft feasibility penalties from realized next-period returns through $C_m(q)$. The learning target then discounts net returns and penalizes drawdown and volatility. All penalty weights are selected on validation records and frozen before the trading holdout:

$$r_{t+1,m}^{\text{net}} = \left(h_{t+1|t,m}^* \right)^\top r_{t+1,m}^{\text{asset}} - C_m(q_{t+1,m}), \quad (4a)$$

$$J_{\text{net}}^{(m)} = \mathbb{E} \left[\sum_{t=0}^T \gamma^t \left(r_{t+1,m}^{\text{net}} - \lambda_{\text{dd}} \text{DD}_{t,m} - \lambda_{\text{vol}} \sigma_{t,m} \right) \right]. \quad (4b)$$

This closes the two-level decision process: asset-level observations determine expert targets, but only synchronized market holdings determine reward, risk, and the reported portfolio metrics.

4 MM-ARC Framework

4.1 Multimodal Market Representation

The representation stage asks what information is available at the decision cutoff and why multiple views are useful. Qwen3-VL-8B-Instruct encodes aligned chart, 100-observation numerical, and structured technical-summary views derived from the same OHLCV record [1]. These views expose visual patterns, numerical scale, and technical semantics without introducing an independent information source.

We use 4-bit QLoRA for parameter-efficient adaptation [5, 9]. Supervised regime learning produces the prior $p_{t,u}$ from training records; its labels are not deployment inputs. The encoder, schema generator, and optional fidelity evaluator share only the base checkpoint, not adapters, prompts, output heads, or calibration records. Appendix A specifies view construction and label timing, and Appendix B documents task isolation.

4.2 Adaptive Capital Routing

The router determines how much capital each interpretable expert receives. Equation (2) combines the fused market state with the SFT prior and the admitted pools' robustness evidence, producing continuous simplex weights rather than selecting one expert. The

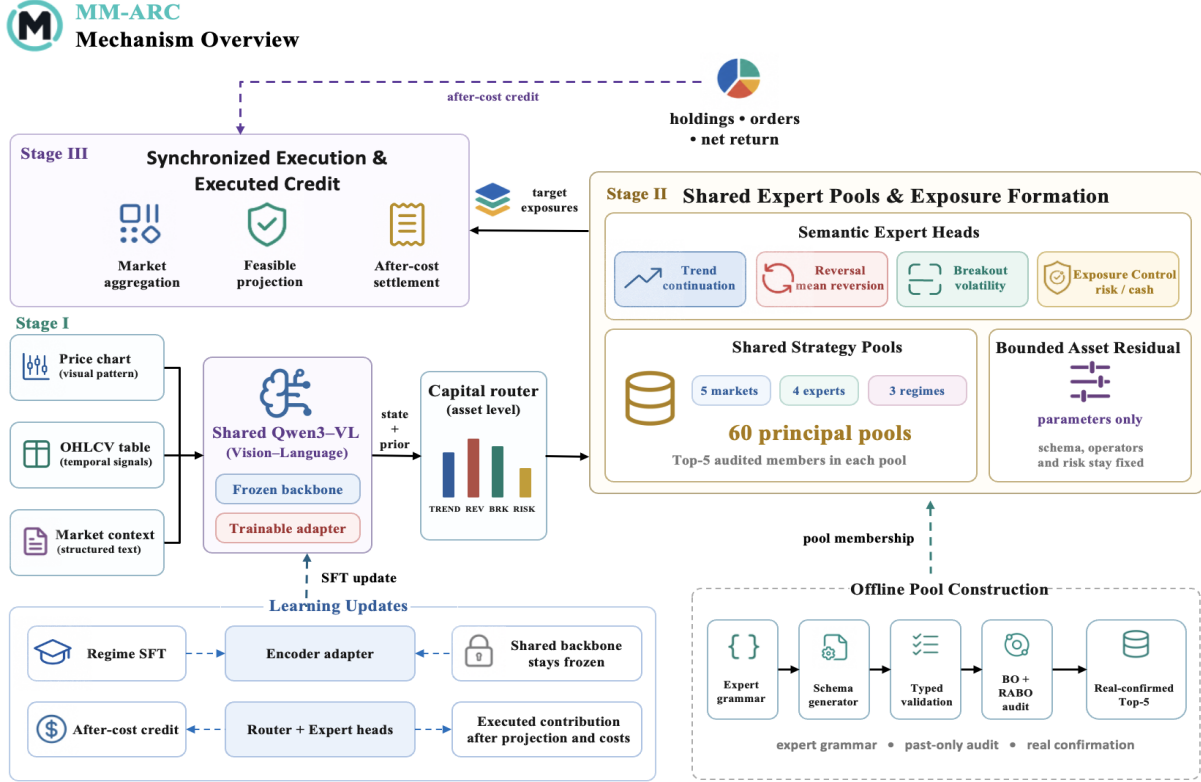


Figure 1: Overview of MM-ARC. The solid path maps each asset’s observed multimodal history to a fused state, continuous expert-capital weights, target exposures, and market-level portfolio construction. Dashed arrows denote robustness weighting and training feedback rather than online information flow.

prior stabilizes state conditioning, while lower-tail evidence tilts capital toward pools that remain reliable on past validation blocks.

After SFT, the backbone is frozen and the adapters, router, and expert heads are optimized with post-projection, after-cost actor-critic reward. Entropy regularization discourages collapse, a prior penalty stabilizes the transition from SFT, and a weight-turnover penalty discourages unnecessary reallocations. Five independent runs use seeds 42–46, with checkpoints selected on validation records. Full training and executed-credit definitions appear in Appendix E.

4.3 Expert Strategy Pools and Asset Adaptation

The four experts encode trend, reversal, breakout, and exposure control. Candidates use typed, expert-specific schemas instead of free-form executable code; Appendix C gives their grammar and validation fields.

For market m , expert k , and regime r , the shared market–expert–regime strategy pool is

$$\mathcal{P}_{m,k,r} = \{(s_j, \bar{\theta}_j, c_j^{\text{rob}})\}_{j=1}^{M_{\text{pool}}}, \quad M_{\text{pool}} = 5. \quad (5)$$

Five markets, four experts, and three regimes yield 60 shared pools rather than $62 \times 4 \times 3 = 744$. For asset u , the bounded asset-specific

parameter adjustment (residual) is

$$\theta_{u,j} = \Pi_{\Theta_{S_j}}(\bar{\theta}_j + D_{S_j} \delta_{u,j}), \quad \|\delta_{u,j}\|_{\infty} \leq \epsilon_{\theta}, \quad (6)$$

It can change only active numeric parameters, never operators, entry/exit logic, or risk constraints. Zero-centered regularization and declared bounds preserve shared strategy semantics; fitting uses only training data or historically visible validation blocks.

4.4 Robustness-Audited Strategy Admission

Admission asks whether a candidate is reliable across past validation blocks, not whether it attains the single highest validation return. Each candidate $x = (s, \theta)$ is therefore evaluated on synchronized tradable market portfolios:

$$J_{\text{pool}}(x; \mathcal{B}_b, m, k, r) = J_{\text{net}}(\text{Port}_m(x, k, r; \mathcal{B}_b)), \quad (7)$$

$$D_{m,k,r}(x) = \{J_{\text{pool}}(x; \mathcal{B}_b, m, k, r)\}_{b=1}^{B_{m,r}}.$$

Here Port_m is the tradable market portfolio after applying the common projection and cost ledger. Appendix C gives the leave-one-asset influence diagnostic and sparse-cell shrinkage rule.

Bayesian optimization proposes schema–parameter pairs within the declared grammar. RABO evaluates their past-only distributions and combines five within-update ranks:

$$S_{\text{RABO}}(x_i) = .30R_P + .30R_{\mathcal{Q}_5^{\text{cand, val}}} + .15R_{\text{med}} + .15R_{\text{stab}} - .10R_{\text{turn}}, \quad (8)$$

The terms rank matched-block benchmark exceedance, the validation 5% quantile, median behavior, local parameter stability, and realized turnover. The five highest audited candidates enter the pool. Appendix D provides the reproducible benchmark, stability neighborhood, controls, and DF-RABO efficiency study.

Selection and routing use related but distinct quantities. Equation (8) ranks candidates within one update; after admission, the pool-level robustness score for expert k is

$$c_{m,r,k}^{\text{rob}} = \sigma \left(\frac{Q_5(D_{m,k,r}^{\text{pool}}) - q_{\min}}{\tau_g} \right) P_{D_{m,k,r}^{\text{pool}}} (J_{\text{net}} > J_{\text{bench}}) S_{m,k,r}^{\text{pool}}, \quad (9)$$

Here J_{bench} is the matched-block after-cost score of the predeclared fixed-seed strategy reference for the same market, expert, regime, and validation block. The reference uses the same synchronized portfolio construction and cost ledger, is fixed before BO and RABO, and is never reselected candidate by candidate. S^{pool} is mean local stability, while $q_{\min}$ and τ_g are validation-frozen. Thus an admitted pool with deteriorating lower-tail evidence is down-weighted rather than treated as equally reliable.

Under Online-WF, pool updates occur every 20 trading days or after a predeclared persistent-regime or validation-decay trigger. Every update at τ uses only observations through τ . During the frozen trading holdout used for the main tables, schema generation, search, pool membership, adapters, router, and expert heads are all fixed before the holdout boundary.

4.5 Market-Feasible Portfolio Construction

Expert targets become orders only after synchronized, market-specific portfolio construction. Equations (3a)–(3b) aggregate the asset-level expert targets and project them onto position, leverage, short-sale, cash, and turnover constraints. All trainable methods and compatible baselines use this same layer; A-shares additionally impose long-only feasibility.

At decision time t , the system uses records available through the market close and submits the projected order at the next tradable open. Equation (4a) settles realized holdings and applicable costs, so reward and reported performance depend only on executed portfolios. Training attribution uses post-projection exposure and realized turnover cost; exact reconciliation, leave-one-expert-out diagnostics, the QP, and priority clipping are specified in Appendices E and F.

5 Experimental Setup

We organize the evaluation around four questions: **RQ1**, how MM-ARC’s equal-market after-cost risk–return performance compares with the baselines, together with market-level heterogeneity relative to the passive benchmark; **RQ2**, whether adaptive capital routing, aligned modalities, and expert specialization contribute beyond matched controls; **RQ3**, whether RABO improves lower-tail behavior relative to peak-based BO selection; and **RQ4**, whether market-shared pools reduce measured search workload without losing portfolio quality.

5.1 Observed Market Data and Temporal Protocol

The benchmark contains 62 real instruments: 10 A-shares, 15 U.S. equities, 15 ETFs, 20 futures, and Bitcoin and Ethereum. Core-market histories begin in January 2017; cryptocurrency histories begin in January 2021; all records end on June 30, 2026. The temporal modality contains OHLCV and technical indicators, the visual modality contains charts rendered from the same causal window, and the text modality contains deterministic technical summaries, regime descriptions, support/resistance levels, and volume analysis. Asset-specific calendars are retained.

The split follows one global calendar. Training ends in December 2023, validation spans January 2024 through June 2025, and the 90-observation labeling horizon is purged at split boundaries. Table 1 distinguishes the one-year frozen trading holdout from the shorter regime-label evaluation. Throughout the holdout, all learned and selected states remain fixed; only forward inference, order generation, and ex post scoring are permitted. Regime-classification metrics end on January 31, 2026 (January 30 for exchange-traded sessions) because they require a 90-session forward label. Trading evaluation requires only next-period settlement and therefore continues through June 30, 2026. Every input, supervisory label, and candidate score carries a cutoff timestamp, and any source time later than its decision time is rejected.

The normalized 10-bps and market-specific 18/8/6/5/25-bps cost protocols are fixed before model selection and are never selected using validation performance.

Corporate actions are handled before feature construction; futures use lagged dominant-contract stitching and a separate roll ledger; cryptocurrency retains its 24/7 calendar. Appendix A reports data provenance, membership, missingness, and cutoff checks.

5.2 Baselines, Training, and Fair Comparison

The comparison covers Buy-and-Hold, a TPE-searched technical baseline, PPO, Time-VLM-style forecasting followed by the common allocation layer, FinAgent (adapted), and LLMoE-style routing [7, 13, 15, 24, 25]. FinAgent and LLMoE-style are protocol-adapted local implementations because their native universes, information interfaces, and portfolio layers are not directly interchangeable with ours. This labeling avoids treating a local reproduction as the original paper’s reported result.

The controls are chosen to isolate successive capabilities. Buy-and-Hold fixes the observed market path and supplies a passive reference. Stochastic technical search uses seeds 42–46 to initialize independent TPE-based BO runs with identical predeclared rules, parameter domains, validation objective, and fixed budget, then freezes each validation winner for the holdout; its variation thus reflects optimizer randomness, not different market histories. PPO tests continuous exposure control without the multimodal shared-pool architecture. Time-VLM-style maps aligned multimodal forecasts to target positions but lacks regime-conditioned shared pools. FinAgent (adapted) retains an agent-style decision interface under the common data and portfolio setting. LLMoE-style is the closest routing control: it uses four routed experts but removes the complete shared-pool and robustness-audit mechanism. The ablations



RABO Pool Construction and Refresh Protocol

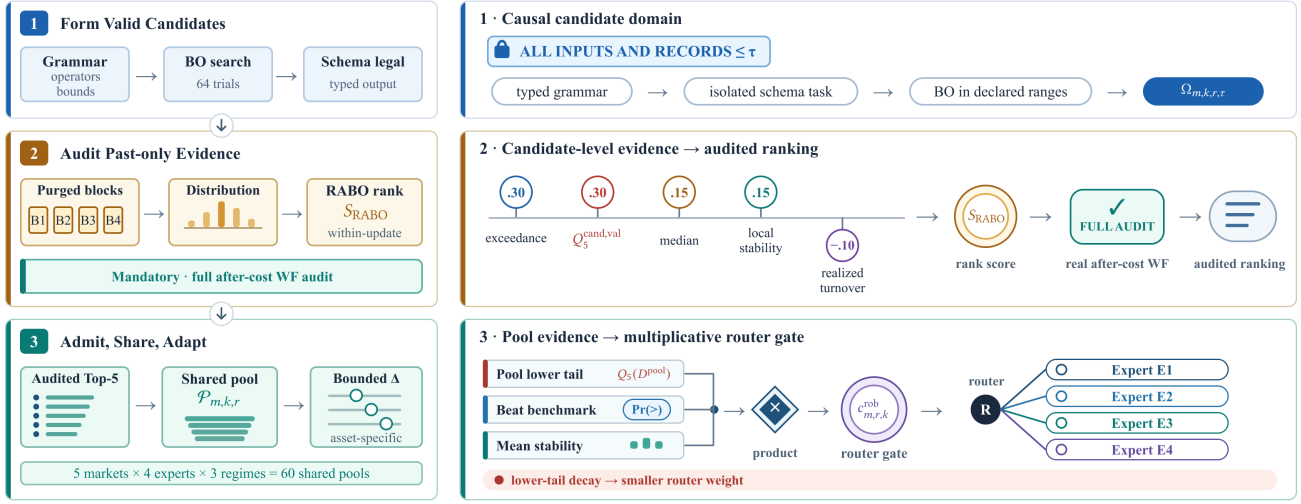


Figure 2: RABO pool construction and robustness-gated routing. At causal cutoff τ , typed grammars and BO generate legal candidates; purged past-only blocks provide the five ranks in S_{RABO} , and a full after-cost walk-forward audit admits the top five to shared market–expert–regime pools with bounded asset residuals. Pool lower-tail quality, benchmark exceedance, and mean stability are multiplied into $c_{m,r,k}^{\text{rob}}$, which scales the corresponding expert’s router weight.

Table 1: Temporal protocol and observed multi-market benchmark. Session counts cover the complete July 2025–June 2026 frozen trading holdout for each training seed.

A. Phase	Dates	Permitted use			
Training	Core: Jan. 2017–Dec. 2023; crypto: Jan. 2021–Dec. 2023	Model and base-strategy fitting			
Validation	Jan. 2024–Jun. 2025	Checkpoint, RABO-weight, Top- K , refresh-threshold, and model-hyperparameter selection; 90-observation boundary purge			
Frozen trading holdout	Jul. 1, 2025–Jun. 30, 2026	Forward inference, orders, and ex post scoring only; all learned and selected states fixed			
Regime-label evaluation	Jul. 1, 2025–Jan. 31, 2026	Classification metrics requiring the 90-session forward label; Jan. 30 is the last exchange-traded session			
Online-WF (supplementary)	Same initial state and trading period	Past-only causal updates; reported separately from the holdout result			
B. Market	Instruments	Sessions/seed	Annualization	Exposure support	Applicable cost components
A-shares	10	242	252	[0, .95]	commission, slippage, sell-side stamp duty
U.S. equities	15	251	252	[−.95, .95]	commission, slippage
ETFs	15	251	252	[−.95, .95]	commission, slippage
Futures	20	251	252	[−.95, .95]	commission, slippage, roll
Cryptocurrency	2	365	365	[−.95, .95]	commission, slippage, funding

additionally replace adaptive capital weights with uniform allocation, remove or corrupt one modality, flatten the expert structure, remove exposure control, and exchange RABO for peak-, quantile-, or turnover-based admission.

To isolate state dependence from the benefit of merely learning a non-uniform expert mixture, we prespecify a learned-static-weight control for each training seed. It retains the same encoder, four experts, strategy pools, portfolio projection, cost ledger, validation objective, turnover constraint, and search budget as the full router, but optimizes one fixed simplex vector using validation records only:

$$\begin{aligned}
 \mathbf{w}_{\text{static}}^{(s)} &= \arg \max_{\mathbf{w} \in \Delta^4} J_{\text{val}}^{(s)}(\mathbf{w}), \\
 \mathbf{w}_{t,u}^{(s)} &= \mathbf{w}_{\text{static}}^{(s)} \quad \text{for every holdout } t, u.
 \end{aligned} \tag{10}$$

During the holdout this control does not read $z_{t,u}$, the regime prior, or robustness-gate evidence. Its weights are never inferred

from holdout averages. The adaptive-routing claim is evaluated against this control rather than against uniform weights or a single validation-best expert alone.

Every trainable method uses the same training/validation cutoffs, tradable universe, rebalance clock, and five seeds. Every method that emits a target position is passed through the same market-feasibility projection and cost accounting. Buy-and-Hold uses the common observed benchmark path and is therefore identical across training seeds. Checkpoint selection, BO/RABO weights, active-pool size, refresh thresholds, and all ablation settings are determined on training and validation records and frozen before the trading holdout.

All model and strategy selection ends before the holdout: checkpoints never use holdout targets, candidates are evaluated only on purged validation blocks, and reporting uses fixed prediction and order records. For each learned method, a seed identifies a complete independent training run rather than a resampling of the

same return path. Five-seed dispersion therefore measures optimization stability, while the block bootstrap separately captures serial dependence in the held-out portfolio record.

5.3 Execution, Metrics, and Statistical Inference

All main results use one normalized, all-in charge on executed one-way turnover:

$$C_{t,m}^{\text{norm}} = c_{\text{norm}} \|q_{t,m}\|_1, \quad c_{\text{norm}} = 10^{-3} = 10 \text{ bps}, \quad (11)$$

where $\|q_{t,m}\|_1$ is the absolute executed change as a fraction of market-portfolio NAV. Hence 100% one-way turnover costs 0.10% of NAV; buying a fully invested position and later liquidating it costs approximately 0.20% in total. Commission, slippage, sell-side stamp duty, roll cost, and funding fields allocate this single charge for attribution and are not added on top. For sensitivity analysis, predictions, gross returns, and orders are held fixed while c_{norm} changes over $\{0, 5, 10, 20, 50\}$ bps. The separate market-specific setting replaces Equation (11), rather than supplementing it, with effective rates of 18/8/6/5/25 bps for A-shares, U.S. equities, ETFs, futures, and cryptocurrency.

We compute total return (TR), Sharpe ratio (SR), maximum draw-down (MDD), annualized one-way turnover, and $Q_{5,20d}^{\text{port}}$, the 5th percentile of rolling 20-session after-cost portfolio returns. The daily risk-free rate is zero; exchange-traded markets use 252 sessions per year and cryptocurrency uses 365. Annualized one-way turnover is $A_m T_m^{-1} \sum_t \|q_{t,m}\|_1$. Equations (23)–(26) give the remaining ledger-level definitions and the equal-market reporting operator. Metrics are first computed for each seed–market portfolio and then equally averaged over the five markets. Mean and seed SD summarize five matched runs. The primary method-difference CI uses 1,200 draws from a 10-session market-stratified circular-block bootstrap applied to matched seed–market–time–block differences; method-level marginal CIs are retained in Appendix G. White’s Reality Check and Hansen’s SPA test address family-level data snooping. Holm-adjusted two-sided t -tests pair the five equal-market seed aggregates and are reported only as auxiliary optimization-stability evidence in the appendix.

The evidence criteria are fixed before reading the holdout results. RQ1 requires an equal-market comparison against all baselines together with explicit market-level heterogeneity and failure-case reporting for MM-ARC and the passive benchmark, rather than one favorable point estimate. RQ2 requires matched routing, modality, and expert controls with identical dates and settlement. RQ3 requires comparing BO and RABO on the same portfolio-level lower-tail variable, together with return and drawdown. RQ4 requires completed-job counts and hardware logs, not pool-count arithmetic alone. A favorable paired test is not used to override an unfavorable SPA or Reality Check result.

Appendices A and H document the evaluation protocol, ledger schemas, and reconciliation definitions.

6 Results and Analysis

6.1 Main Cross-Market Performance

Table 2 reports the one-year frozen trading holdout under the normalized all-in 10-bps protocol. Equal-market averages five separately settled portfolio metrics; MDD is not a pooled-curve draw-down. MM-ARC records TR 14.1%, SR 1.33, and MDD -13.7% , compared with 10.4%, .53, and -18.3% for the closest routing baseline, LLMoE-style. The corresponding effect sizes are .80 in SR and 4.6 percentage points in MDD, while annualized one-way turnover falls from $12.5\times$ to $7.2\times$.

The market-level results remain heterogeneous. U.S. Buy-and-Hold has higher raw TR than MM-ARC (32.3% versus 24.2%), although MM-ARC records SR 1.55 and MDD -16.5% there. Cryptocurrency is MM-ARC’s weakest market portfolio (SR .94; MDD -20.6%). Retaining this failure case in the equal-market aggregate makes the evidence about risk-adjusted performance and draw-down control rather than universal raw-return dominance.

Table 3 reports matched method differences and their paired block-bootstrap intervals. Against LLMoE-style, $\Delta\text{SR} = +.80$ with CI $[.26, 1.46]$; the risk-effect point estimates are $\Delta\text{MDD} = +4.6$ and $\Delta Q_{5,20d}^{\text{port}} = +2.15$ percentage points. Against global learned-static weights, the corresponding differences are $+21$, $+1.6$, and $+1.38$, with the SR CI $[.10, .39]$. Against market-specific learned-static weights, $\Delta\text{SR} = +.12$ with CI $[.06, .21]$. Under the prespecified benchmark and candidate-family construction, SPA ($p = .039$) and Reality Check ($p = .021$) reject their family-level nulls. The family-level evidence therefore supports benchmark-relative superiority within the evaluated family and holdout, but not dominance over every candidate or future-regime generalization. Block-length results appear in Appendix G.

6.2 Routing, Modalities, and Expert Structure

Panel A of Table 4 answers RQ2. The router records TR 14.1%, SR 1.33, MDD -13.7% , and $Q_5 -8.22\%$; global static gives 13.5%, 1.12, -15.3% , and -9.60% , while market-specific static gives 13.8%, 1.21, -14.5% , and -8.85% . Uniform and validation-best weights reach SR .88 and .98; removing or corrupting the visual view lowers SR to 1.14 and .85.

Panel C evaluates expert decomposition. A flat pool reaches SR .90, and removing exposure control deepens MDD from -13.7% to -18.2% . Thus, the exposure expert is not merely an interpretability label: it materially changes the executed risk profile. The benefit of the full four-expert model over the flat-pool control also supports the use of semantic restrictions during strategy search.

6.3 RABO and Shared-Pool Analysis

Panel D answers RQ3. BO-best delivers the highest TR (15.0%) but has $Q_{5,20d}^{\text{port}} = -11.85\%$ and MDD -17.3% . Full RABO gives -8.22% and -13.7% at TR 14.1%. The Q_5 -only rule is marginally better on the designated quantile (-8.20%), but lower on SR (1.14 versus 1.33) and worse on MDD (-14.2% versus -13.7%). The composite audit therefore trades 0.9 percentage points of TR relative to the validation-peak selector for a broader improvement in lower-tail and drawdown behavior.

Table 2: Equal-market results over the July 2025–June 2026 frozen trading holdout under the normalized 10-bps (0.10%) one-way all-in cost. Values are mean $\pm$ sample SD over the five matched runs in Table 16: training seeds for learned methods and TPE/BO seeds for stochastic technical search.

Method	TR% (Mean $\pm$ Seed SD)	SR (Mean $\pm$ Seed SD)	MDD% (Mean $\pm$ Seed SD)
B&H	14.0 $\pm$ 0.0	0.58 $\pm$ 0.00	-24.7 $\pm$ 0.0
Stoch. technical search	8.9 $\pm$ 0.9	0.45 $\pm$ 0.03	-20.3 $\pm$ 0.6
PPO	8.4 $\pm$ 0.6	0.45 $\pm$ 0.06	-18.3 $\pm$ 0.3
Time-VLM-style	9.7 $\pm$ 0.9	0.49 $\pm$ 0.06	-18.5 $\pm$ 0.5
FinAgent (adapted)	8.7 $\pm$ 0.8	0.44 $\pm$ 0.03	-20.6 $\pm$ 0.3
LLMoE-style	10.4 $\pm$ 0.9	0.53 $\pm$ 0.05	-18.3 $\pm$ 0.4
MM-ARC	14.1 $\pm$ 0.9	1.33 $\pm$ 0.06	-13.7 $\pm$ 0.2

Table 3: Difference-focused evidence on the frozen holdout. Effects are the first-named method minus the control; positive Δ MDD and $\Delta Q_{5,20d}^{\text{port}}$ are favorable. The 95% CIs are obtained by resampling matched seed–market–time–block differences. SPA and Reality Check are family-level diagnostics shown once.

Comparison	Δ SR	95% CI	Δ MDD (pp)	95% CI	$\Delta Q_{5,20d}^{\text{port}}$ (pp)	95% CI	SPA p	Reality Check p
MM-ARC – LLMoE-style	+0.80	[0.26, 1.46]	+4.6	[2.03, 6.94]	+2.15	[.78, 3.56]	.039	.021
MM-ARC – global learned static	+0.21	[0.10, 0.39]	+1.6	[.25, 2.70]	+1.38	[.30, 2.57]	—	—
MM-ARC – market-specific learned static	+0.12	[.06, .21]	+0.8	[.05, 1.32]	+0.63	[.15, 1.14]	—	—
Full RABO – BO-best	+0.31	[.13, .55]	+3.6	[1.36, 5.68]	+3.63	[1.64, 5.65]	—	—

Table 4: Five-seed mean ablations on the frozen holdout. Metrics are equal-market summaries and $Q_{5,20d}^{\text{port}}$ is the common rolling-20-session lower-tail statistic. Global learned-static weights follow Eq. (10).

Panel	Variant	TR%	SR	MDD%	$Q_{5,20d}^{\text{port}}\%$
A: Routing	Regime-tilted router	14.1	1.33	-13.7	-8.22
	Global learned static weights	13.5	1.12	-15.3	-9.60
	Market-specific learned static	13.8	1.21	-14.5	-8.85
	Uniform weights	11.3	0.88	-16.6	-11.60
	Validation-best expert	12.0	0.98	-17.6	-12.00
B: Modalities	Full aligned views	14.1	1.33	-13.7	-8.22
	Temporal + text	12.6	1.14	-15.3	-10.91
	Corrupted visual	10.9	0.85	-17.2	-11.35
C: Experts	Four semantic experts	14.1	1.33	-13.7	-8.22
	Flat pool	11.1	0.90	-17.3	-11.62
	No exposure control	13.0	1.10	-18.2	-13.64
D: Pools/search	Fixed seed pool	10.2	0.79	-18.4	-11.61
	BO-best	15.0	1.02	-17.3	-11.85
	Turnover-matched BO	14.3	0.99	-16.2	-9.87
	Q5-only audit	13.2	1.14	-14.2	-8.20
	Equal-weight audit	13.6	1.20	-14.5	-9.59
	Asset-specific RABO	13.9	1.26	-13.9	-8.57
	Shared, no residual	13.4	1.22	-14.2	-9.36
	Shared + residual (full)	14.1	1.33	-13.7	-8.22

The shared-plus-residual design reaches SR 1.33, compared with 1.26 for asset-specific RABO and 1.22 for sharing without residuals. Relative to asset-specific RABO, its point differences are +0.2 percentage points in TR, +.07 in SR, +0.2 percentage points in MDD, and +0.35 percentage points in $Q_{5,20d}^{\text{port}}$. Table 5 reports the matched-block intervals for these differences. All four intervals cross zero, so the comparison does not establish a quality difference between the shared and asset-specific designs. The observed pattern is consistent with market-level sharing providing common evidence while bounded asset-specific adjustments preserve useful heterogeneity; workload reduction is evaluated separately below.

6.4 Mechanism, Workload, and Cost Diagnostics

Figure 3(a) compares BO-best and RABO on the same rolling after-cost portfolio variable. RABO shifts Q5 rightward while leaving

Table 5: Shared + residual minus asset-specific RABO. Positive differences favor the shared design. Confidence intervals use the same 1,200-draw matched seed–market–time–block circular-block bootstrap as the primary comparisons.

Metric	Point difference	Paired-block 95% CI
TR (percentage points)	+0.2	[−1.4, +2.0]
SR	+0.07	[−0.20, +0.37]
MDD (percentage points)	+0.2	[−1.3, +1.5]
$Q_{5,20d}^{\text{port}}$ (percentage points)	+0.35	[−0.9, +1.5]

the median slightly lower, whereas BO-best retains higher full-period TR. In panel (b), trend peaks in uptrends (.29), reversal in downtrends (.32), and exposure control in sideways periods (.25). Temporal weight variance is .00192, pairwise symmetric KL divergence is .00677, and entropy is 1.363. These modest but systematic state-dependent tilts are consistent with adaptive capital routing

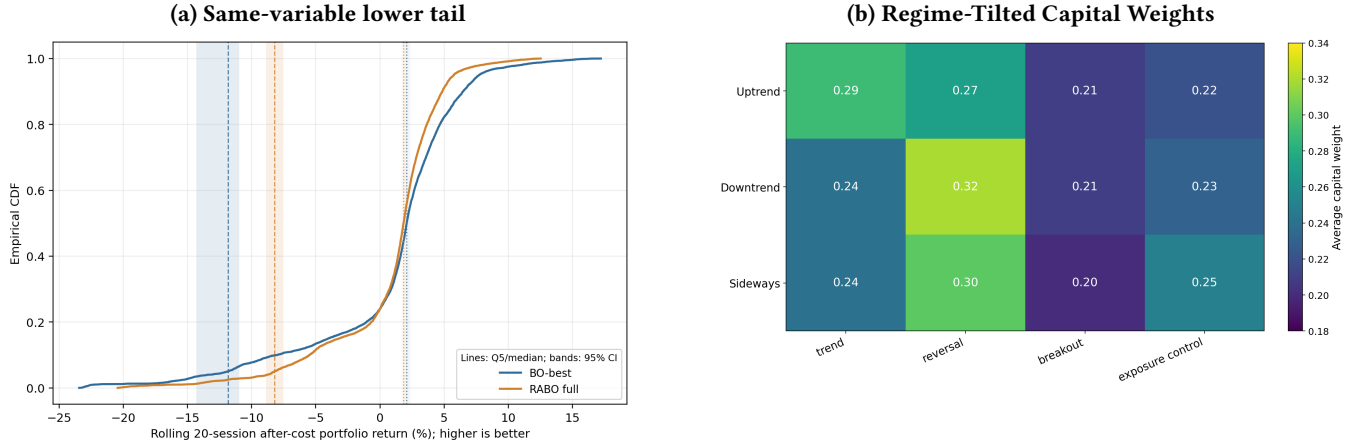


Figure 3: Mechanism diagnostics. Panel (a) marks Q5 with dashed lines and the median with dotted lines; shaded bands are 95% seed-market cluster-bootstrap intervals. Panel (b) summarizes capital weights over the three regimes that index the shared pools.

within a diversified expert mixture, rather than hard expert selection or discrete regime switching.

For RQ4, structural sharing reduces the pool count from 744 asset-specific pools to 60 market-expert-regime pools. The measured workload also falls: complete backtests decline from 43,192 to 3,594, and end-to-end time from 31.6 to 4.6 hours, including 0.42 hours of residual fitting. CPU time falls from 126.4 to 18.7 hours, GPU time from 18.2 to 3.2 hours, and peak memory from 28.4 to 21.8 GB under the same environment. The smaller proportional wall-clock gain relative to the pool-count reduction reflects fixed orchestration, data loading, and residual-fitting overheads.

Holding predictions, gross returns, and orders fixed, increasing the normalized all-in rate from 10 to 50 bps reduces MM-ARC TR from 14.1% to 10.8% and SR from 1.33 to 1.00; LLMoE-style falls from 10.4% to 4.8% and from .53 to .24. Under the replacement market-specific 18/8/6/5/25-bps schedule, MM-ARC records TR 13.9% and SR 1.31, versus LLMoE-style SR 0.50. The qualitative comparison is therefore not tied to the normalized 10-bps point.

6.5 Research-Question Synthesis

The four analyses support a qualified conclusion. RQ1 favors MM-ARC in risk-adjusted point estimates and paired intervals, and both family-level tests reject their prespecified nulls, although the market-level results retain meaningful heterogeneity. RQ2’s matched controls favor adaptive routing over both learned-static alternatives and support the modalities and architecture. RQ3 shows a 0.9-point TR trade-off for shallower MDD and a less negative Q5. RQ4 shows lower pool count and measured work. These findings connect architectural choices to executed outcomes without relying on one headline metric.

7 Discussion and Limitations

Temporal scope. The July 2025–June 2026 frozen trading holdout is one historical realization and cannot establish adaptation across longer market cycles. In the retained cryptocurrency failure, seed 46 loses 20.5% between January 18 and February 5, 2026, showing that positive full-period TR and SR can coexist with a concentrated loss

when a fast volatility transition outpaces the frozen pool. Longer forward paper trading and prospectively declared causal refresh checks are therefore required.

Statistical scope. SPA ($p = .039$) and Reality Check ($p = .021$) reject their family-level nulls under the prespecified benchmark, candidate set, and performance differential. These results strengthen the benchmark-relative evidence after data-snooping correction on the observed holdout, but they do not show that MM-ARC dominates every candidate or that the advantage generalizes to future regimes. Five seeds measure optimization variability, not uncertainty over alternative market histories; the block bootstrap captures dependence within the same holdout rather than variation across future regimes.

Universe and information scope. The 62 instruments across five asset classes do not represent the complete investable universe, and survivorship bias may remain where point-in-time membership is unavailable. The text view is derived from market records, so the study does not test independent news, filings, or macroeconomic reasoning. Larger and more heterogeneous universes may also require sector- or liquidity-level parents rather than one shared hierarchy per market.

Execution and capacity scope. The common ledger models commissions, slippage, stamp duty, roll cost, funding, position caps, and turnover, but not every queue, borrow, outage, capacity, or nonlinear market-impact effect. Capacity-aware simulation and forward execution are needed before interpreting the observed results as deployable scale.

8 Conclusion

On this frozen holdout, MM-ARC yields favorable risk-adjusted and drawdown estimates relative to the closest routing baseline and both learned-static controls, and SPA and Reality Check reject their prespecified family-level nulls. Retained ablations and workload measurements are consistent with contributions from adaptive routing, the visual view, exposure control, RABO, and shared-pool workload reduction. Together, these results support a benchmark-relative advantage within the evaluated candidate family and period,

Table 6: Measured search workload under a fixed hardware and software environment. Counts include completed jobs; time and peak memory come from the recorded resource logs.

Design	Gen.	Valid	BO prop.	Backtests	Residual h	E2E h	CPU h	GPU h	Peak GB
Asset-specific RABO	4,464	3,578	47,616	43,192	0.00	31.6	126.4	18.2	28.4
Shared + residual	360	291	3,840	3,594	0.42	4.6	18.7	3.2	21.8

Measurement environment. EPYC 7543P/256 GB/A100 40 GB; Python 3.11, PyTorch 2.7.1/CUDA 12.8; four FIFO workers, batch 1, cache off, failures/retries included; E2E includes all stages through table write-out. One-second psutil/nvidia-smi sampling; three-run E2E ranges: 30.9–32.4 h and 4.4–4.8 h.

without claiming dominance over every candidate or future-regime superiority.

References

- [1] Shuai Bai et al. 2025. Qwen3-VL Technical Report. arXiv:2511.21631
- [2] David H. Bailey, Jonathan M. Borwein, Marcos Lopez de Prado, and Qiji Jim Zhu. 2016. The Probability of Backtest Overfitting. *Journal of Computational Finance* 20, 4 (2016), 39–69.
- [3] David H. Bailey and Marcos Lopez de Prado. 2014. The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality. *The Journal of Portfolio Management* 40, 5 (2014), 94–107.
- [4] Zhuo Chen, Xinzhe Yuan, Jianshu Zhang, Jinzong Dong, Ruichen Zhou, Yingchun Niu, Tianhang Zhou, Yu Yang Fredrik Liu, Yuqiang Li, Nanyang Ye, and Qinying Gu. 2026. LABO: LLM-Accelerated Bayesian Optimization through Broad Exploration and Selective Experimentation. Proceedings of the 43rd International Conference on Machine Learning.
- [5] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient Finetuning of Quantized LLMs. In *Advances in Neural Information Processing Systems*, Vol. 36.
- [6] Qianggang Ding, Haochen Shi, Jiadong Guo, and Bang Liu. 2024. Tradexpert: Revolutionizing trading with mixture of expert llms. arXiv:2411.00782
- [7] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. 2018. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 80)*. PMLR, Stockholm, Sweden, 1861–1870.
- [8] Peter Reinhard Hansen. 2005. A Test for Superior Predictive Ability. *Journal of Business & Economic Statistics* 23, 4 (2005), 365–380.
- [9] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. LoRA: Low-Rank Adaptation of Large Language Models. *arXiv preprint arXiv:2106.09685* (2021).
- [10] Ming Jin, Shiyu Wang, Lintao Ma, Zhixuan Chu, James Zhang, Xiaoming Shi, Pin-Yu Chen, Yuxuan Liang, Yuan-fang Li, Shirui Pan, et al. 2024. Time-LLM: Time Series Forecasting by Reprogramming Large Language Models. The Twelfth International Conference on Learning Representations. <https://openreview.net/forum?id=Unb5CVPtat>
- [11] Marc C. Kennedy and Anthony O'Hagan. 2000. Predicting the Output from a Complex Computer Code When Fast Approximations Are Available. *Biometrika* 87, 1 (2000), 1–13. doi:10.1093/biomet/87.1.1
- [12] Sang Il Lee and Seong Joon Yoo. 2019. Multimodal Deep Learning for Finance: Integrating and Forecasting International Stock Markets. arXiv:1903.06478
- [13] Kuan-Ming Liu and Ming-Chih Lo. 2025. LLM-Based Routing in Mixture of Experts: A Novel Framework for Trading. arXiv:2501.09636
- [14] Robert Pardo. 2008. *The Evaluation and Optimization of Trading Strategies* (2nd ed.). John Wiley & Sons.
- [15] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. arXiv:1707.06347
- [16] Jasper Snoek, Hugo Larochelle, and Ryan P. Adams. 2012. Practical Bayesian Optimization of Machine Learning Algorithms. In *Advances in Neural Information Processing Systems*, Vol. 25. Curran Associates, Inc., Lake Tahoe, Nevada, USA, 2951–2959.
- [17] Dave Walton. 2014. Know Your System! Turning Data Mining from Bias to Benefit through System Parameter Permutation. *2014 NAAIM Wagner Award Winner* (2014). doi:10.2139/ssrn.2423187
- [18] Halbert White. 2000. A Reality Check for Data Snooping. *Econometrica* 68, 5 (2000), 1097–1126.
- [19] Sheng Wu, Ozan Irsoy, Shuohang Lu, Valentin Dabravolski, Mark Dredze, Sebastian Gehrmann, Prabhajan Kambadur, David Rosenberg, and Gideon Mann. 2023. BloombergGPT: A Large Language Model for Finance. arXiv:2303.17564
- [20] Yijia Xiao, Edward Sun, Di Luo, and Wei Wang. 2024. TradingAgents: Multi-agents LLM financial trading framework. arXiv:2412.20138
- [21] Qianqian Xie, Weiguang Han, Xiao Zhang, Yanzhao Lai, Min Peng, Alejandro Lopez-Lira, and Jimin Huang. 2023. Pixiu: A large language model, instruction data and evaluation benchmark for finance. arXiv:2306.05443
- [22] Huicheng Yang, Xiao-Yang Liu, and Chenguang Wang. 2023. FinGPT: Open-Source Financial Large Language Models. arXiv:2306.06031
- [23] Yuan Yu, Hao Li, Zhixin Chen, Yuxuan Jiang, Yingjie Li, and Dong Zhang. 2024. FinMem: A Performance-Enhanced LLM Trading Agent with Layered Memory and Character Design. AAAI Spring Symposium on AI in FinTech.
- [24] Wei Zhang, Lei Zhao, Hao Xia, Shiyu Sun, Meng Qin, and Xiang Li. 2024. A Multimodal Foundation Agent for Financial Trading: Tool-Augmented, Diversified, and Generalist. Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining.
- [25] Siru Zhong, Weilin Ruan, Ming Jin, Huan Li, Qingsong Wen, and Yuxuan Liang. 2025. Time-VLM: Exploring Multimodal Vision-Language Models for Augmented Time Series Forecasting. arXiv:2502.04395

Anonymous Executable Pipeline and Experiment Results

Open repository ↗

anonymous.4open.science/r/MM-ARC-32F7

A Temporal Evaluation and Data Provenance

A.1 Calendar and Frozen Evaluation

The benchmark is constructed from point-in-time market records. Core-market histories span January 2017 through June 30, 2026, and cryptocurrency histories span January 2021 through the same cutoff. Core-market training uses January 2017 through December 2023, cryptocurrency training begins in January 2021, and validation spans January 2024 through June 2025. The frozen trading holdout runs from July 1, 2025 through June 30, 2026. A-shares, U.S. equities, ETFs, and futures follow their native exchange calendars, while cryptocurrency uses a continuous daily calendar; returns are not forward-filled across closed markets.

Training, validation, BO, RABO, schema formation, and checkpoint selection are completed before the trading holdout is opened. Throughout the full year, the encoder adapters, router, expert heads, shared pools, asset-specific parameter adjustments, and execution configuration remain fixed. Regime-classification metrics are reported through January 31, 2026 (January 30 for exchange-traded sessions) because their labels require 90 future sessions; after-cost trading metrics extend through June 30 because they require only next-period settlement. The supplementary Online-WF protocol begins from the same initial state, is reported separately, and restricts every update at τ to inputs, labels, validation blocks, and candidate evaluations available no later than τ .

A.2 Artifact Integrity and Leakage Controls

The executable pipeline applies provenance and leakage checks to its inputs and processed samples. For each available input artifact, the validator records its source, retrieval timestamp, adjustment convention, first and last observation, row count, missingness, duplicate count, and SHA-256 digest. Processed multimodal samples additionally carry the instrument, decision timestamp, maximum source timestamp, feature version, chart-rendering version, and split identifier. A sample fails validation if any source timestamp is later than its decision time or if its target timestamp is visible to training, schema formation, BO, RABO, or checkpoint selection.

Portfolio ledgers produced by the pipeline carry method, training seed, market, date, protocol phase, previous holding, requested target, executed holding, order, rejected notional, gross return, component costs, and net return. Instrument records retain stable market identifiers and point-in-time universe membership. These schemas support record-level reconciliation of predictions, orders, holdings, costs, and reported metrics during pipeline execution.

A.3 Multimodal View and Label Construction

At each decision time, the sample builder extracts the 100 observations ending at the recorded cutoff. The visual branch renders a standard candlestick chart and a CYC-MRKAB chart containing moving averages, Bollinger Bands, MACD, RSI, and KDJ. The temporal branch contains OHLCV and technical indicators normalized

with training-only statistics. The structured-text branch deterministically summarizes returns, volatility, moving-average ordering, indicator positions, support/resistance, and volume from the same window. The three branches are generated separately and then joined by instrument and decision time; none adds independent news, filings, or macroeconomic records.

SFT targets use the designated 90-session forward regime horizon only within the labeling phase. Samples whose target horizon crosses a split boundary are purged, and target timestamps are inaccessible to feature construction, strategy search, and checkpoint selection. These labels supervise and evaluate the regime prior $p_{t,u}$ but are never supplied as contemporaneous inference inputs.

B Qwen3-VL Roles and Training Isolation

The implementation uses three task-separated Qwen3-VL roles. They share the Qwen3-VL-8B-Instruct base checkpoint but not adapters, output heads, prompts, or task-specific training state. The execution evaluator is deterministic and does not use a language-model judgment.

The encoder is first adapted with regime supervision and then used by the after-cost actor-critic stage. The generator is trained to emit a typed schema that passes expert-specific grammar, indicator, parameter-range, and risk-control checks. The fidelity evaluator is calibrated only on past paired evaluations. No evaluator score can substitute for J_R when a candidate enters the Top-5 active pool.

C Shared Pools and Search-Space Accounting

C.1 Pool Hierarchy

The shared pool $\mathcal{P}_{m,k,r}$ reuses strategy structure and base parameters for one market, expert type, and regime. Equation (6) limits each asset to bounded, schema-valid continuous or integer offsets (residuals). Categorical operators, entry/exit logic, and risk constraints remain shared. Residual clipping uses the tighter of the schema range and a predeclared relative radius; zero-centered initialization and an ℓ_2 penalty discourage unnecessary personalization.

Here G is the number of valid schemas considered per pool and $N_{\text{trial}} = 64$ is the BO trial budget. The reduction in shared strategy pools and active members is exact. End-to-end measurements in Table 6 additionally include schema validation, data loading, orchestration, residual fitting, and evaluation overhead.

Candidate influence is checked through

$$\Delta_{u,b}(x) = J_{\text{pool}}(x; \mathcal{B}_b, m, k, r) - J_{\text{pool}}^{(-u)}(x; \mathcal{B}_b, m, k, r), \quad (12)$$

where the second term removes asset u from the same block and renormalizes the remaining assets in that market portfolio. We record the complete influence distribution and the largest absolute asset share before admitting a market-shared candidate.

To compare residual magnitudes across heterogeneous parameter scales, let $\mathcal{A}_{s_j}$ be the active numeric parameters of schema s_j and define

$$\rho_{u,j} = \left[\frac{1}{|\mathcal{A}_{s_j}|} \sum_{d \in \mathcal{A}_{s_j}} \left(\frac{(D_{s_j} \delta_{u,j})_d}{\theta_{s_j,d}^{\max} - \theta_{s_j,d}^{\min}} \right)^2 \right]^{1/2}. \quad (13)$$

A component is counted as boundary-saturated when either the residual bound or the projected schema bound is active within the

Table 7: Benchmark composition and operational handling. Every row is evaluated on observed market records under its native calendar.

Market	Instruments	Evaluation sessions/seed	Position and settlement	Data handling
A-shares	10	242	long-only, T+1, sell-side stamp duty	adjusted prices, suspensions, exchange calendar
U.S. equities	15	251	bounded long/short, margin limits	corporate actions, borrow and exchange-calendar checks
ETFs	15	251	venue-dependent shorting and cash limits	distributions and underlying-asset classification
Futures	20	251	contract leverage, margin, next-open rolls	lagged dominant contract and explicit roll ledger
Cryptocurrency	2	365	continuous trading and bounded leverage	exchange-source, funding, and outage checks
Total	62	—	five separately settled market portfolios	equal-market reporting after portfolio-level metrics

Table 8: Task separation in the implemented model. Only full walk-forward evaluation can authorize pool admission.

Role	Input/output	Isolation	Authority
Multimodal encoder	charts, series, summary $\rightarrow z_{t,u}, p_{t,u}$	encoder adapter and regime head	representation and routing only
Schema generator	grammar and past context $\rightarrow$ typed schema	generator adapter, prompt, and decoding seed	candidate proposal only
Fidelity evaluator	schema and past context $\rightarrow J_L$	evaluator adapter and calibrated regression head	acquisition priority only
Execution evaluator	executable schema $\rightarrow J_R$	fixed walk-forward engine and validation protocol	mandatory evidence for admission

Table 9: Structural search-space accounting. The reduction is not an end-to-end speedup.

Quantity	Asset-specific	Market-shared	Reduction
Shared strategy pools	744	60	91.9%
Active members	3,720	300	91.9%
Candidate budget	$744GN_{\text{trial}}$	$60GN_{\text{trial}}$	91.9%

audit tolerance. Low, middle, and high residual-magnitude groups are validation-defined tertiles of $\rho_{u,j}$ and are frozen before holdout scoring.

Table 10: Prespecified residual and influence diagnostics. Magnitude groups are validation-defined residual-norm tertiles frozen before holdout scoring.

Item	Reported statistic	Value
<i>A. Residual and influence distribution</i>		
Residual norm ρ	median / P90 / maximum	0.05 / 0.14 / 0.21
Boundary saturation	fraction of active components clipped	10%
Boundary saturation breakdown	clipped fraction by schema and market	schema 6–14%; market 11/9/8/10/12%
Leave-one-asset influence	P95 / maximum $ \Delta_{u,b} $	0.08 / 0.21
<i>B. Holdout performance by residual magnitude</i>		
Low tertile	assets / TR / SR / MDD	20 / 11% / 1.2 / –14%
Middle tertile	assets / TR / SR / MDD	21 / 14% / 0.9 / –11%
High tertile	assets / TR / SR / MDD	21 / 13% / 1.1 / –13%
High-tertile concentration	top-asset share / volatility-adjusted check	14% / SR 1.03; MDD –13.2%

Let $D_{m,k,r}(x)$ denote the regime-specific candidate distribution and $D_{m,k}(x)$ its market–expert parent. With $B_{\min} = 20$ effective

blocks, sparse-cell shrinkage is

$$\begin{aligned} \tilde{D}_{m,k,r}(x) &= \omega_{m,r} D_{m,k,r}(x) + (1 - \omega_{m,r}) D_{m,k}(x), \\ \omega_{m,r} &= \min\left(1, \frac{B_{m,r}}{B_{\min}}\right). \end{aligned} \quad (14)$$

The effective-block count, shrinkage weight, and pre/post-shrinkage quantiles are retained for every update. This avoids ranking a sparse regime cell by an unstable empirical 5% quantile.

C.2 Schema Fields

D RABO and DF-RABO Details

D.1 Robustness Score and Controls

For candidate set $\Omega_{m,k,r,\tau}$, every component in Eq. (8) is ranked within the same pool update. R_P ranks $P(J_{\text{pool}} > J_{\text{bench}})$; $R_{Q_5^{\text{cand, val}}}$ ranks the 5% quantile of the shrunk validation distribution; R_{med} ranks its median; R_{stab} measures performance retention under local parameter perturbations; and R_{turn} ranks realized turnover. Rank normalization removes incompatible scales without converting all negative lower-tail outcomes to the same clipped value.

Let $x_{m,k,r}^{\text{seed}}$ denote the predeclared fixed-seed strategy-pool reference for market m , expert k , and regime r . Its matched-block benchmark is

$$J_{\text{bench},b}^{(m,k,r)} = J_{\text{pool}}\left(x_{m,k,r}^{\text{seed}}; \mathcal{B}_b, m, k, r\right). \quad (15)$$

Table 11: Machine-readable strategy schema and validation constraints.

Field	Constraint	Example role
expert	trend/reversal/breakout/exposure	fixes the admissible grammar
indicators	declared list with bounded causal windows	MA, RSI, ATR, volume
entry/exit	typed Boolean expression tree	crossover or threshold rule
position	bounded target-exposure rule	long/flat/short or target volatility
risk	stop, leverage, cash, and turnover bounds	execution-safe constraints
parameters	typed range, integer flag, and default	search domain Θ_s
provenance	adapter, prompt, seed, source cutoff	reproducibility and leakage audit

For candidate x , benchmark exceedance is the fraction of the same validation blocks on which its after-cost objective is larger:

$$P(J_{\text{pool}} > J_{\text{bench}}) = \frac{1}{B_{m,r}} \sum_{b=1}^{B_{m,r}} \mathbb{I}[J_{\text{pool}}(x; \mathcal{B}_b, m, k, r) > J_{\text{bench},b}^{(m,k,r)}]. \quad (16)$$

The reference is fixed before BO and RABO, uses the same synchronized portfolio construction and cost ledger as the candidate, and is never reselected candidate by candidate.

For candidate $x = (s, \theta)$, define its scalar local-validation objective as the median after-cost outcome over the matched blocks:

$$J_{\text{val}}(s, \theta) = \text{median}_{1 \leq b \leq B_{m,r}} J_{\text{pool}}((s, \theta); \mathcal{B}_b, m, k, r). \quad (17)$$

Let $\mathcal{A}_s$ index the active numeric parameters of schema s , e_d be the unit vector for dimension d , and $[\theta_d^{\min}, \theta_d^{\max}]$ its declared search range. The one-at-a-time 5% neighborhood is

$$\mathcal{N}(\theta) = \left(\bigcup_{d \in \mathcal{A}_s} \left\{ \Pi_{\Theta_s} \left(\theta \pm .05(\theta_d^{\max} - \theta_d^{\min})e_d \right) \right\} \right) \setminus \{\theta\}. \quad (18)$$

Projection enforces the declared bounds; integer parameters are rounded to the nearest legal integer, projected duplicates are replaced by the nearest distinct legal neighbor when one exists, and categorical operators remain fixed. After duplicate neighbors are removed, local stability is

$$\text{Stability}(\theta) = \text{clip}_{[0,1]} \left(1 - \text{median}_{\theta' \in \mathcal{N}(\theta)} \frac{|J_{\text{val}}(s, \theta') - J_{\text{val}}(s, \theta)|}{|J_{\text{val}}(s, \theta)| + \varepsilon_f} \right). \quad (19)$$

Algorithm 1 Market-shared RABO pool update

Require: market m , expert k , regime r , causal cutoff τ

- 1: Form grammar-valid schemas using records visible by τ
 - 2: Run BO over each declared parameter space
 - 3: **for** candidate x_i **do**
 - 4: Evaluate synchronized $J_{\text{pool}}(x_i)$ on purged validation blocks
 - 5: Apply Eq. (14) when the regime cell is sparse
 - 6: Compute exceedance, $Q_5^{\text{cand, val}}$, median, stability, and turnover ranks
 - 7: Compute $S_{\text{RABO}}(x_i)$ using Eq. (8)
 - 8: **end for**
 - 9: Fully evaluate every candidate eligible for admission
 - 10: Admit the five highest audited candidates and fit bounded asset-specific parameter adjustments
 - 11: **return** $\mathcal{P}_{m,k,r}$ and pool-level robustness evidence $c_{m,r,k}^{\text{rob}}$
-

D.2 Mixed Representation and Fidelity Gate

The evaluator encoder produces a frozen schema embedding $e_s \in \mathbb{R}^{d_s}$. For the union of expert-specific parameter domains, $\tilde{\theta} \in [0, 1]^{d_\theta}$ stores normalized values and $a_s \in \{0, 1\}^{d_\theta}$ marks active dimensions. Inactive values are ignored by the numerical distance, integer dimensions are rounded only before execution, and categorical choices use separate embeddings.

The semantic and masked numerical kernels are combined with a paired Kennedy–O’Hagan fidelity model [11]:

$$k(x, x') = k_{\text{sem}}(e_s, e_{s'}) k_{\text{num}}([\tilde{\theta}; a_s], [\tilde{\theta}'; a_{s'}]), \quad (20)$$

$$J_R(x) = \rho J_L(x) + \delta(x), \quad p_\delta(x) = \frac{\sigma_\delta^2(x)}{\rho^2 \sigma_L^2(x) + \sigma_\delta^2(x)}.$$

Candidates with high discrepancy dominance receive priority for full evaluation; every admission candidate still receives the same complete after-cost walk-forward audit. Table 13 reports a separate budget-matched six-method search diagnostic comparing vanilla BO, full-fidelity RABO search, LLM-initialized BO, evaluator-score-only search, random-fidelity RABO, and DF-RABO. Its metrics belong to that diagnostic benchmark and are not directly comparable to the end-to-end equal-market holdout ablation in Table 4.

E Training and Executed-Credit Accounting

The first training stage minimizes regime cross-entropy on training records. The second stage freezes the backbone and optimizes the router, expert heads, and execution adapters with clipped actor-critic updates. The reward includes after-cost return and penalties for drawdown, turnover, constraint projection, and abrupt router changes. The validation objective and all penalty weights are fixed before the frozen trading holdout.

For synchronized raw expert vector $v_{t+1|t,m,k}$, define smoothed sign-specific shares for each asset component u :

$$a_{t+1|t,u,k}^\pm = \frac{[\pm v_{t+1|t,m,k,u}]_+ + \epsilon/K}{\sum_j [\pm v_{t+1|t,m,j,u}]_+ + \epsilon}. \quad (21)$$

The executed allocation is

$$\tilde{h}_{t+1|t,m,k,u} = \begin{cases} a_{t+1|t,u,k}^+ h_{t+1|t,m,u}^* & h_{t+1|t,m,u}^* \geq 0, \\ a_{t+1|t,u,k}^- h_{t+1|t,m,u}^* & h_{t+1|t,m,u}^* < 0. \end{cases} \quad (22)$$

Because $\sum_k a_{t+1|t,u,k}^\pm = 1$, expert allocations reconcile exactly with the executed portfolio. Cost is allocated by each expert’s post-projection turnover, not its unexecuted raw target. The leave-one-expert-out diagnostic independently recomputes projection and settlement without expert k and compares the counterfactual change with the allocated contribution.

Table 12: Five-seed mean RABO-control results on the frozen holdout under the matched execution protocol.

Control	Admission rule	TR%	SR	MDD%	$Q_{5,20d}^{\text{port}}\%$	Turn.
BO-best	highest validation peak only	15.0	1.02	-17.3	-11.85	9.8
Median-only	median of $\bar{D}_{m,k,r}(x)$	13.5	1.18	-14.8	-9.10	8.0
Lower-tail-only	$Q_5^{\text{cand, val}}$; CVaR sensitivity retained	13.2	1.14	-14.2	-8.20	7.5
Stability-only	local parameter-neighborhood stability	12.9	1.10	-14.6	-9.35	7.0
Turnover-matched BO	BO-best after matching realized turnover to RABO	14.3	.99	-16.2	-9.87	7.3
Equal-weight audit	equal signed weights for the five RABO components	13.6	1.20	-14.5	-9.59	7.6
Full RABO	Eq. (8) with validation-frozen weights	14.1	1.33	-13.7	-8.22	7.2
Weight sensitivity A	(.35, .25, .15, .15, -.10)	14.0	1.31	-14.0	-8.35	7.3
Weight sensitivity B	(.25, .35, .15, .15, -.10)	13.9	1.32	-13.9	-8.15	7.1
Weight sensitivity C	(.30, .30, .20, .10, -.10)	14.2	1.29	-14.1	-8.44	7.4

Table 13: Independent dual-fidelity search diagnostic under matched full-evaluation budgets. Its metrics are not directly comparable to the end-to-end equal-market holdout ablation in Table 4. Elapsed time is measured in the same environment and normalized to vanilla BO.

Method	SR	MDD%	$Q_{5,20d}^{\text{port}}\%$	Full backtests	Evaluator calls	Relative time
Vanilla BO	.69	-17.2	-4.31	64	0	1.00×
Full-fidelity RABO search	.81	-13.8	-2.74	64	0	1.07×
LLM-initialized BO	.74	-16.0	-3.65	64	20	1.05×
Evaluator-score-only	.40	-21.5	-5.92	0	200	.18×
Random-fidelity RABO	.57	-17.0	-4.10	28	200	.55×
DF-RABO	.82	-13.6	-2.69	28	200	.56×

F Execution Rules and Additional Evaluation

F.1 Projection and Priority Clipping

Equation (3b) is a convex projection with an ℓ_1 turnover term; an exact epigraph-form QP solution is retained as an audit reference. The implemented fast path applies the same predeclared priority order to every compatible method. Each stage can only reduce the requested exposure, and the final ledger rejects any holding that fails the complete feasible-set check.

Algorithm 2 Market-level priority clipping

Require: raw target $h_{t+1|t,m}^{\text{raw}}$, prior holding $h_{t,m}^*$, rules C_m

- 1: Remove unavailable orders and enforce long/short eligibility
- 2: Clip each asset to its position bound
- 3: Enforce gross/net exposure and leverage caps
- 4: Enforce the required cash buffer
- 5: Clip $q = h - h_{t,m}^*$ to the turnover limit
- 6: Recheck C_m ; reduce exposure if needed
- 7: **return** feasible $h_{t+1|t,m}^*$ and order $q_{t+1,m}$

At date t , the model uses records available through the market close and submits $q_{t+1,m} = h_{t+1|t,m}^* - h_{t,m}^*$ at the next tradable open. Suspended or unavailable assets retain their prior feasible holding until an order can be executed. The primary normalized ledger applies Equation (11) as one all-in charge. Its component fields allocate that total for attribution only. In the market-specific evaluation, commission, slippage, stamp duty, roll cost, and funding rules replace—and never supplement—the normalized charge; A-share stamp duty is nonzero only for executed sells.

F.2 Metric Definitions and Equal-Market Aggregation

For one fixed training seed, let $r_{t,m}^{\text{net}}$ be the daily after-cost return of market portfolio m over its T_m native sessions; the seed index is suppressed. The daily risk-free rate is fixed at zero. Define $A_m =$

252 for exchange-traded business-day markets and $A_m = 365$ for cryptocurrency. With

$$\begin{aligned}\bar{r}_m^{\text{net}} &= \frac{1}{T_m} \sum_{t=1}^{T_m} r_{t,m}^{\text{net}}, \\ (s_m^{\text{net}})^2 &= \frac{1}{T_m - 1} \sum_{t=1}^{T_m} (r_{t,m}^{\text{net}} - \bar{r}_m^{\text{net}})^2, \\ \text{SR}_m &= \sqrt{A_m} \frac{\bar{r}_m^{\text{net}}}{s_m^{\text{net}}},\end{aligned}\quad (23)$$

the reported Sharpe ratio is therefore computed from the market portfolio, not by averaging asset-level Sharpe ratios. Let $V_{0,m} = 1$ and $V_{t,m} = \prod_{\tau=1}^t (1 + r_{\tau,m}^{\text{net}})$. Maximum drawdown is

$$\text{MDD}_m = \min_{1 \leq t \leq T_m} \left(\frac{V_{t,m}}{\max_{0 \leq s \leq t} V_{s,m}} - 1 \right) \leq 0, \quad (24)$$

and is reported as a negative percentage. Annualized one-way turnover uses the same executed orders as the cost ledger:

$$\text{Turn}_m = \frac{A_m}{T_m} \sum_{t=1}^{T_m} \|q_{t,m}\|_1. \quad (25)$$

Finally, after computing any reported metric g_m on each market’s native calendar, the equal-market summary is

$$g^{\text{eq}} = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} g_m = \frac{1}{5} \sum_{m \in \mathcal{M}} g_m. \quad (26)$$

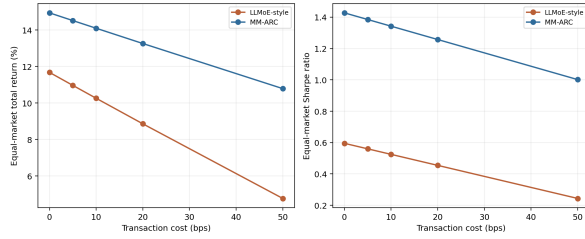
This is a reporting operator over five separately settled market portfolios, not a globally synchronized fund.

G Complete Numeric Diagnostics

Table 16 reports all five matched equal-market runs used for the main mean and SD. Learned methods use training seeds, stochastic technical search uses TPE/BO seeds, and Buy-and-Hold is invariant. Table 17 retains auxiliary seed dispersion, marginal intervals, turnover, and Holm tests separately from Table 3. Table 18 reports

Table 14: Market-specific feasibility, cost components, and operational handling.

Market	Position feasibility	Rebalance/calendar	Applicable cost components	Additional handling
A-shares	long-only, asset/gross caps	exchange days, T+1	commission, sell stamp duty, slippage	adjusted prices and suspensions
U.S. equities	bounded long/short, margin	exchange calendar	commission, spread/impact proxy	corporate actions and borrow limits
ETFs	venue-dependent long/short	venue calendar	commission and slippage	distributions and underlying class
Futures	contract leverage and margin	lagged dominant contract; next-open roll	fees, slippage, roll cost	multipliers and roll ledger
Cryptocurrency	bounded leverage	continuous 24/7	fees, slippage, funding	exchange source and outage checks

**Figure 4: Transaction-cost sensitivity over 0, 5, 10, 20, and 50 bps. Predictions, gross returns, and orders remain fixed; only the charged friction changes.****Table 15: Market-specific friction sensitivity. This setting replaces the normalized charge; predictions, gross returns, and orders are fixed. Rates are predeclared execution assumptions, and entries are five-seed means.**

Setting	Cost bps	MM-ARC			LLMoE-style		
		TR%	SR	MDD%	TR%	SR	MDD%
A-shares	18	15.0	1.40	-12.6	12.2	0.64	-16.7
U.S. Equities	8	24.7	1.55	-16.6	19.8	0.68	-22.2
ETFs	6	15.1	1.50	-9.8	10.6	0.58	-15.2
Futures	5	9.6	1.37	-9.6	6.4	0.49	-13.4
Crypto	25	5.0	0.74	-20.3	0.4	0.10	-22.7
Equal-market	mixed	13.9	1.31	-13.8	9.9	0.50	-18.0

the 10/20/40-session path-level sensitivity analysis, and Table 19 exposes the per-market results. Tables 20 and 21 report unresampled regime support and record-level checks.

Table 16: Equal-market results for five matched runs under the normalized 10-bps (0.10%) one-way all-in cost. Each cell is TR% / SR / MDD%. Seeds index training runs for learned methods and TPE/BO runs for stochastic technical search; B&H is identical because neither changes its observed market path.

Method	Seed 42	Seed 43	Seed 44	Seed 45	Seed 46
B&H	14.0/0.58/-24.7	14.0/0.58/-24.7	14.0/0.58/-24.7	14.0/0.58/-24.7	14.0/0.58/-24.7
Stoch. technical search	7.6/0.49/-20.2	9.9/0.44/-19.8	8.9/0.41/-20.3	9.5/0.47/-19.9	8.8/0.44/-21.2
PPO	7.9/0.39/-18.6	9.2/0.54/-18.2	7.8/0.39/-17.9	8.7/0.47/-18.3	8.2/0.44/-18.3
Time-VLM-style	8.6/0.45/-18.5	10.8/0.59/-18.4	9.2/0.50/-19.3	10.2/0.49/-17.8	9.7/0.44/-18.7
FinAgent (adapted)	7.5/0.41/-20.7	9.3/0.49/-20.4	8.4/0.42/-20.8	9.5/0.46/-20.2	8.8/0.43/-20.8
LLMoE-style	9.2/0.46/-18.3	11.4/0.59/-17.8	9.8/0.50/-18.9	10.7/0.55/-18.2	10.8/0.56/-18.2
MM-ARC	13.0/1.28/-13.9	15.1/1.33/-13.8	13.4/1.28/-13.9	14.8/1.36/-13.7	14.2/1.41/-13.3

Table 17: Optimization-stability and marginal method summaries for the frozen trading holdout. Mean and Seed SD summarize five independently trained equal-market seed aggregates. Marginal time-block CIs describe each method separately and are not substitutes for the paired difference CI in Table 3. Holm p values are auxiliary optimization-stability evidence from two-sided tests pairing the five seed aggregates against MM-ARC.

Method	Mean SR	SR Seed SD	Marginal block 95% CI (SR)	Mean MDD%	MDD Seed SD	One-way turnover ($\times$ /yr)	Holm p
PPO	0.45	0.06	[-0.03, 1.31]	-18.3	0.3	11.1	< .001
FinAgent (adapted)	0.44	0.03	[-0.01, 0.94]	-20.6	0.3	17.5	< .001
Time-VLM-style	0.49	0.06	[0.02, 1.17]	-18.5	0.5	13.6	< .001
LLMoE-style	0.53	0.05	[0.08, 1.19]	-18.3	0.4	12.5	< .001
MM-ARC	1.33	0.06	[0.87, 2.69]	-13.7	0.2	7.2	—

Table 18: Block-length sensitivity for paired path-level inference. Each metric cell reports the point difference followed by its paired 95% CI from the 1,200-draw market-stratified circular-block bootstrap.

Comparison	Block	Δ SR [95% CI]	Δ MDD (pp) [95% CI]	$\Delta Q_{5,20d}^{\text{port}}$ (pp) [95% CI]
MM-ARC – LLMoE-style	10	+0.80 [0.26, 1.46]	+4.6 [2.03, 6.94]	+2.15 [.78, 3.56]
	20	+0.80 [.13, 1.52]	+4.6 [1.88, 7.15]	+2.15 [.47, 3.85]
	40	+0.80 [.14, 1.59]	+4.6 [1.24, 7.79]	+2.15 [.33, 4.07]
MM-ARC – market-specific static	10	+0.12 [.06, .21]	+0.8 [.05, 1.32]	+0.63 [.15, 1.14]
	20	+0.12 [.02, .26]	+0.8 [.11, 1.42]	+0.63 [.03, 1.33]
	40	+0.12 [.07, .33]	+0.8 [.15, 1.52]	+0.63 [.04, 1.44]
Full RABO – BO-best	10	+0.31 [.13, .55]	+3.6 [1.36, 5.68]	+3.63 [1.64, 5.65]
	20	+0.31 [.03, .62]	+3.6 [1.16, 5.88]	+3.63 [1.21, 6.07]
	40	+0.31 [.06, .73]	+3.6 [.62, 6.37]	+3.63 [.84, 6.53]

Table 19: Per-market five-seed mean results under the normalized 10-bps protocol. Each cell is TR% / SR / MDD%, computed from the retained method–seed–market daily-return records. Market cells are rounded separately; equal-market summaries are computed from the unrounded seed–market records and can therefore differ by one displayed rounding unit from an average of the displayed cells.

Method	A-shares	U.S. equities	ETFs	Futures	Crypto
Buy-and-Hold	19.8/.72/-22.0	32.3/.92/-29.1	16.1/.75/-16.7	4.6/.35/-18.2	-2.9/.18/-37.6
LLMoE-style	13.3/.69/-16.9	19.7/.68/-22.5	10.4/.56/-15.6	5.7/.44/-13.7	2.9/.28/-22.8
Market-specific learned static	15.2/1.32/-13.3	23.5/1.42/-17.4	14.2/1.30/-10.8	8.8/1.12/-10.7	7.3/.89/-20.3
MM-ARC	15.8/1.46/-12.5	24.2/1.55/-16.5	14.9/1.45/-9.8	9.1/1.30/-9.6	6.5/.94/-20.6

Table 20: Predicted router-regime coverage. Entries are average observed sessions per seed (share of market timestamps).

Setting	Uptrend	Downtrend	Sideways
A-shares	157.8 (65.2%)	38.2 (15.8%)	46.0 (19.0%)
U.S. Equities	151.8 (60.5%)	45.4 (18.1%)	53.8 (21.4%)
ETFs	106.4 (42.4%)	68.6 (27.3%)	76.0 (30.3%)
Futures	118.0 (47.0%)	45.4 (18.1%)	87.6 (34.9%)
Crypto	133.0 (36.4%)	29.2 (8.0%)	202.8 (55.6%)

Table 21: Numerical integrity checks on daily records from the frozen trading holdout.

Identity	Maximum absolute error
Gross return – costs – net return	$< 7 \times 10^{-18}$
Sum of component costs – total cost	$< 7 \times 10^{-18}$
Previous holding + order – executed holding	$< 7 \times 10^{-18}$
Instrument aggregation – market portfolio	$< 6 \times 10^{-17}$

H Result Provenance and Consistency

The anonymous repository contains an executable pipeline and related experiment results for MM-ARC. It covers data processing, training, strategy search, evaluation, and report generation. Full MM-ARC values reconcile across the main results, ablations, RABO comparison, per-seed report, and the normalized all-in 10-bps cost point whenever their aggregation levels match.

Metrics are computed on each seed–market portfolio before equal-market averaging. No asset-average Sharpe ratio is substituted for portfolio Sharpe. Turnover is the absolute change in executed holdings, and costs use the same order ledger that reconciles consecutive holdings. Candidate-level $Q_5^{\text{cand, val}}$ remains confined to past validation blocks, whereas the reported $Q_{5,20d}^{\text{port}}$ is computed from rolling 20-session after-cost returns in the corresponding evaluation record.

The release scope is the executable pipeline and its accompanying experiment results. It does not claim to redistribute the complete upstream market-data corpus or every intermediate record produced during the research process. The favorable paired results and family-level rejections are both reported in this paper.